"Aha, but then the father cleans up the room and thereby reduces the toys' entropy back to 20," you might say. "Doesn't this violate the second law of thermodynamics?" No, not at all. The toys' entropy may be reduced by the father's cleaning, but the entropy in the father's body and in the room's air has increased: It took a lot of energy to throw the toys back onto the northernmost tiles, energy that the father got by "burning up" some of his body's fat. The burning converted neatly organized fat molecules into disorganized waste products, for example, the carbon dioxide that he exhaled randomly into the room; and the resulting increase in the father's and the room's entropy (the increase in the number of ways their atoms and molecules can be distributed) far more than made up for the decrease in the toys' entropy.

Jacob Bekenstein was not persuaded. It seemed likely to him that a black hole's area in some deep sense *is* its entropy—or, more precisely, its entropy multiplied by some constant. If not, Bekenstein reasoned, if black holes have vanishing entropy (no randomness at all) as Hawking claimed, then black holes could be used to decrease the entropy of the Universe and thereby violate the second law of thermodynamics. All one need do is bundle all the air molecules from some room into a small package and drop them into a black hole. The air molecules and all the entropy they carry will disappear from our Universe when the package enters the hole, and if the hole's entropy does not increase to compensate for this loss, then the total entropy of the Universe will have been reduced. This violation of the second law of thermodynamics would be highly unsatisfactory, Bekenstein argued. To preserve the second law, a black hole *must* possess an entropy that goes up when the package falls through its horizon, and the most promising candidate for that entropy, it seemed to Bekenstein, was the hole's surface area.

Not at all, Hawking responded. You can lose air molecules by throwing them down a black hole, and you can also lose entropy. That is just the nature of black holes. We will just have to accept this violation of the second law of thermodynamics, Hawking argued; the properties of black holes require it—and besides, it has no serious consequences at all. For example, although under ordinary circumstances a violation of the second